

PROJECT REPORT
ON
STUDENT GRADE
PREDICTION MODEL

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR
THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE & ENGINEERING/ COMPUTER
SCIENCE & ENGINEERING (CYBER SECURITY) /
MATHEMATICS AND COMPUTING**



Submitted By: Ankit Kumar Ojha

Name:	Priyanshu Gupta	Ankit Kumar Ojha	Gyan Prakash	Krishan Kumar
Roll No.	25BECSE52	25BECSE15	25BECSE32	25BECSE38

(Times New Roman, 10 pt., Bold, Centre)

Group No. 10

Under the supervision of:

Dr. Palak Mahajan

(Assistant Professor)

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
CENTRAL UNIVERSITY OF JAMMU**

Rahya-Suchani (Bagla), District Samba-181143, Jammu (J&K)

CERTIFICATE

This is to certify that the Project work entitled “**Student Grade Prediction Model**” which is submitted by **Ankit Kumar Ojha**(Enrollment No. 25BECSE15), **Gyan Prakash**(Enrollment No. 25BECSE32), **Krishan Kumar**(Enrollment No. 25BECSE38), and **Priyanshu Gupta** x (Enrollment No. 25BECSE52), Semester II , Department of **Computer Science and Engineering, Central University of Jammu**, during the academic year 2025-2029, in partial fulfillment for the award of degree of **Bachelor of Technology**, is a record of the project work carried out by the students under my supervision.

S.No	Student Name	Enrollment No.	Signature
1	Ankit Kumar Ojha	25BECSE15	
2	Gyan Prakash	25BECSE32	
3	Krishan Kumar	25BECSE38	
4	Priyanshu Gupta	25BECSE52	

Place: Central University of Jammu

Date:

To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this university or elsewhere.

Signature

Supervisor Name

ABSTRACT

This project focuses on predicting student academic performance using machine learning techniques. The aim is to analyse various academic and behavioural factors such as study time, previous grades, failures, and absences to predict the final grade of students. A supervised learning approach is implemented using the Random Forest Regression algorithm due to its ability to capture complex relationships within the dataset.

The dataset used for training and testing the model is obtained from Kaggle, consisting of more than 200 labelled student records. Data preprocessing techniques such as handling missing values, removing duplicates, and encoding categorical variables are applied to improve data quality. Exploratory Data Analysis (EDA) is conducted to understand feature relationships and identify key predictors influencing performance.

The model is trained using an 80-20 train-test split and evaluated using performance metrics such as Mean Absolute Error (MAE) and R^2 score. The model achieved an R^2 score of 0.85 and MAE of 1.03, indicating good predictive accuracy.

Additionally, a user-friendly interface is developed using Streamlit to allow real-time prediction of student performance. This system can assist educators in identifying at-risk students and making data-driven academic decisions.

Chapter 1: Introduction

Background

Student performance analysis is an essential aspect of modern education systems. Predicting academic outcomes helps institutions identify weak students early and provide necessary interventions. With the advancement of machine learning, predictive analytics has become a powerful tool in education.

Problem Statement

The objective of this project is to predict student performance (final grade) using machine learning based on academic and behavioral data.

Objectives

- To develop a machine learning model for predicting student performance
- To analyze the impact of various academic factors
- To implement Random Forest Regression for prediction
- To build a user-friendly interface for real-time predictions

Scope

This project can be used in educational institutions to:

- Identify weak students early
- Improve academic planning
- Support data-driven decision-making

Chapter 2: Literature Review / Existing System

Traditional methods of evaluating student performance rely on manual analysis and intuition, which are often inaccurate and inefficient.

Existing systems lack predictive capabilities and fail to utilize historical data effectively.

Recent studies have shown that machine learning models such as Linear Regression, Decision Trees, and Random Forest can significantly improve prediction accuracy. Among these, Random Forest performs better due to its ensemble learning nature, reducing overfitting and improving generalization.

However, many existing systems lack:

- Proper feature selection
- Real-time prediction capability
- User-friendly interfaces

This project addresses these gaps by integrating machine learning with an interactive web application.

Chapter 3: Methodology / System Design

Dataset

- Source: Kaggle
- Records: 200+
- Type: Labelled dataset

Tools & Technologies

- Python
 - Pandas, NumPy
 - Matplotlib, Seaborn
 - Scikit-learn
 - Streamlit
-

System Workflow

1. Data Loading & Preprocessing

- Data loaded from CSV file
 - Removed duplicate records
 - Handled missing values using mean/median
 - Converted categorical variables into numerical form
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2. Exploratory Data Analysis (EDA)

- Visualized grade distribution
 - Analysed relationships between features
 - Identified important predictors (G1, G2, study time)
-

3. Feature Selection & Encoding

Selected features:

- Study time
- Failures
- Absences
- G1, G2

Encoding:

- Gender and other categorical features converted into numerical values
-

4. Train-Test Split

- 80% Training Data
- 20% Testing Data

5. Model Building

Algorithm used: **Random Forest Regression**

Steps:

- Defined input features (X) and target (G3)
- Initialized model
- Trained using training dataset
- Predicted results on test dataset

6. System Architecture

Flow:

Dataset → Preprocessing → EDA → Feature Selection → Model Training → Prediction → Streamlit UI

Chapter 4: Implementation and Results

Implementation

The model is implemented using Python and Scikit-learn.

Code Snippet (Simplified)

```
from sklearn.ensemble import RandomForestRegressor

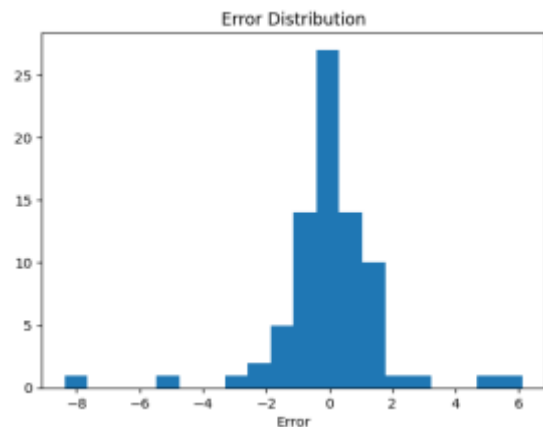
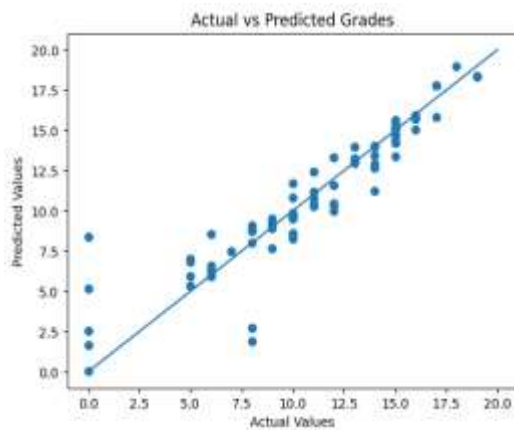
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

Performance Metrics

- **R² Score:** 0.85
- **Mean Absolute Error (MAE):** 1.03

These results indicate strong prediction capability.



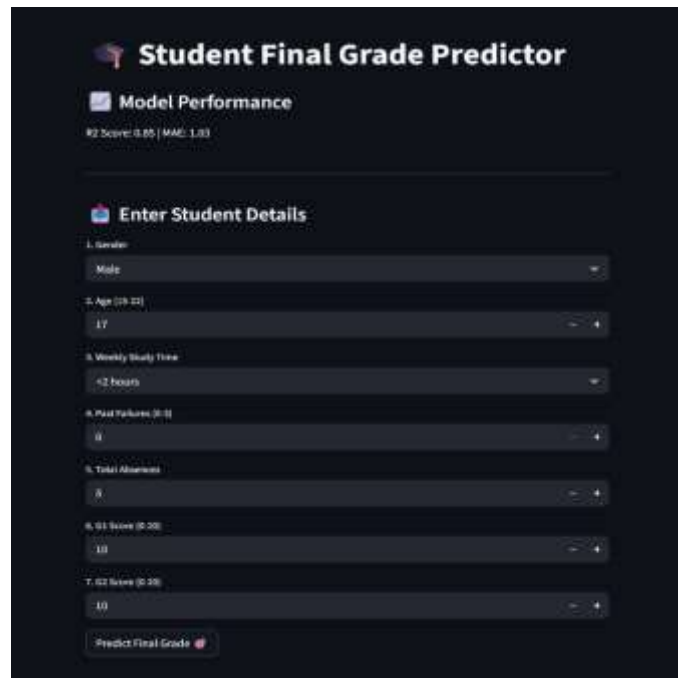
Observations

- G1 and G2 are the most influential features
- Random Forest handles non-linear relationships effectively
- Model performs well on unseen data

System Interface

A Streamlit-based UI allows:

- Input of student data
- Real-time prediction of final grade



The screenshot shows the 'Student Final Grade Predictor' interface. At the top, it displays 'Model Performance' with an R2 Score of 0.85 and MAE of 1.83. Below this is the 'Enter Student Details' section with the following inputs:

- 1. Gender: Male
- 2. Age (19-22): 17
- 3. Weekly Study Time: <2 hours
- 4. Past Failures (0-5): 0
- 5. Total Absences: 8
- 6. G1 Score (0-20): 10
- 7. G2 Score (0-20): 10

A 'Predict Final Grade' button is located at the bottom of the form.



Chapter 5: Conclusion and Future Scope

Conclusion

This project successfully demonstrates the use of machine learning in predicting student performance. The Random Forest Regression model provides accurate predictions and effectively captures relationships between academic factors and final grades.

The integration of a user-friendly interface enhances usability, making it practical for real-world applications. The system can assist educators in identifying students who need additional support.

Future Scope

- Use larger and more diverse datasets
 - Implement deep learning models for improved accuracy
 - Add classification (pass/fail prediction)
 - Deploy as a web application
 - Integrate with school management systems
-

REFERENCES

1. Kaggle Dataset: [Student Performance Dataset](#)
2. Scikit-learn Documentation
3. Python Libraries Documentation (Pandas, NumPy, Matplotlib)